GCSE MATHEMATICS · 2026 EXAM SERIES · FREE BONUS FORMULA SHEET

Calculator Papers — Formula Quick Sheet

The formulas you're most likely to reach for on Papers 2 & 3. A condensed companion to the full formula sheets — focused on the calculator-paper topics. Foundation core + Higher marked.

FOUND + HIGHER needed by everyone **HIGHER** Higher tier only

Why one sheet, not two. Papers 2 and 3 both test the *same* full specification — there is no formula that belongs to Paper 2 but not Paper 3, or vice versa. So a single calculator-papers quick sheet is the honest, useful format. This pulls together the formulas that come up most on the calculator papers; for the complete topic-by-topic formula reference, use the full three-sheet set in this bundle.

1 Trigonometry (calculator essential)

Right-Angled Trig — FOUND + HIGHER

$\sin\theta = \dfrac{opp}{hyp} \cdot \cos\theta = \dfrac{adj}{hyp} \cdot \tan\theta = \dfrac{opp}{adj}$

SOH-CAH-TOA

Find an angle: $\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$

calculator in Degrees mode

Pythagoras: $a^2 + b^2 = c^2$

Sine & Cosine Rules — HIGHER

$\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C}$

sine rule

$a^2 = b^2 + c^2 - 2bc\cos A$

cosine rule

$\text{Area} = \frac{1}{2}ab\sin C$

(these are given on the Higher sheet)

2 Percentages, Interest & Growth

Percentages — FOUND + HIGHER

$\%\ \text{change} = \dfrac{change}{original} \times 100$

Multiplier: *+15% → ×1.15 · −15% → ×0.85*

Reverse %: *divide by the multiplier*

Interest, Growth & Decay — FOUND + HIGHER

Compound: $\text{Value} = P\left(1 + \dfrac{r}{100}\right)^n$

Depreciation/decay: $P\left(1 - \dfrac{r}{100}\right)^n$

Simple interest $= \dfrac{P \times R \times T}{100}$

3 Compound Measures & Number

Compound Measures — FOUND + HIGHER

$\text{Speed} = \dfrac{distance}{time}$

$\text{Density} = \dfrac{mass}{volume}$

$\text{Pressure} = \dfrac{force}{area}$

Standard Form & Bounds — FOUND + HIGHER

$A \times 10^n$ $(1 \le A < 10)$

use the $\times 10^x$ key

Error interval: *value ± half the rounding unit*

Bounds: combine UB/LB depending on operation (Higher)

4 Area, Volume & Circles

Area & Circles

FOUND + HIGHER

Triangle = ½ × base × height

Trapezium = ½(a + b) × h

Circle: $C = \pi d$ · Area = πr^2

Arc = $\frac{\theta}{360}\pi d$ · Sector = $\frac{\theta}{360}\pi r^2$ *(H)*

Volume & Surface Area

FOUND + HIGHER

Prism = cross-section area × length

Cylinder = $\pi r^2 h$

Sphere = $\frac{4}{3}\pi r^3$ · Cone = $\frac{1}{3}\pi r^2 h$ *(H)*

(given on Higher sheet)

Pyramid = $\frac{1}{3}$ × base area × height *(H)*

5 Algebra & Statistics

Algebra

HIGHER

Quadratic formula: $x = \frac{-b \pm \sqrt{(b^2 - 4ac)}}{2a}$

Gradient = $\frac{y_2 - y_1}{x_2 - x_1}$

Line: $y = mx + c$

Statistics

FOUND + HIGHER

Mean = $\frac{\Sigma fx}{\Sigma f}$

use calculator STAT mode

Frequency density = $\frac{\text{frequency}}{\text{class width}}$ *(H)*

IQR = Q3 − Q1 *(H)*

Calculator-paper reminder: Keep your calculator in **Degrees** mode for all trigonometry, carry exact values forward with the **ANS** key instead of rounding early, and only round to the requested accuracy at the very end. See the full **Calculator Skills** guide in this bundle for the button-by-button technique.